

PAPER_348 — Stephan's Quintet Shock Ridge: F_U_Bi_i with KE Density and LENR Energy Coupling

Whitepaper Series: Star-Magic UQFF Phase 2

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Classification: FIRST UQFF F_U_Bi_i for an intergalactic shock ridge with LENR coupling

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Abstract

The complete UQFF buoyancy-unified force $F_{U_Bi_i}$ is computed for the Stephan's Quintet compact group intergalactic shock ridge. The 1500 km/s relative velocity of the NGC 7318b intruder galaxy generates a kinetic energy density $KE_{den} = \frac{1}{2}\rho \cdot \Delta v^2$, which couples to the UQFF vacuum field via FLEENR (Low Energy Nuclear Reaction force component). The shock ridge lies at $x_2 = 290$ Mly and yields $F_{U_Bi_i} \approx -8.32 \times 10^{217}$ N.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force

$$F_{U_Bi_i} \approx -8.32 \times 10^{217} \text{ N}$$

2.2 Shock Kinetic Energy Density

$$KE_{den} = \frac{1}{2}\rho_{IGM} \cdot \Delta v^2 = \frac{1}{2}\rho_{IGM} \cdot (1500 \times 10^3 \text{ m/s})^2$$

where $\rho_{IGM} \approx 10^{-26} \text{ kg/m}^3$ (intragroup medium density at $z \approx 0.021$).

$$KE_{den} \approx \frac{1}{2} \times 10^{-26} \times (1.5 \times 10^6)^2 = 1.125 \times 10^{-14} \text{ J/m}^3$$

2.3 LENR Energy Coupling (Kozima/FLEENR)

The UQFF includes a Low Energy Nuclear Reaction force component:

$$E_{FLEENR} = E_{Kozima} \cdot \frac{\rho_{UA}}{\rho_{SCm}} \cdot [SSq]$$

where E_{Kozima} represents nuclear binding energy release in dense shock-compressed plasma.

2.4 Cross-System Separation

$$x_2 = 290 \text{ Mly} = 2.90 \times 10^{23} \text{ m}$$

Distance from observer to the Stephan's Quintet shock ridge (Hickson Compact Group HCG 92).

3. Key Values

Quantity	Formula	Value
Δv	Relative shock velocity	1500 km/s
KE_den	$\frac{1}{2}\rho\cdot\Delta v^2$	$\sim 10^{-14}$ J/m ³
F_U_Bi_i	UQFF full 5-eq	-8.32×10^{217} N
x_2	Distance (observer)	290 Mly
E_FLENR	LENR coupling	Kozima $\times$ [SSq] $\times$ ρ_{ratio}
ρ_{IGM}	Intragroup medium	$\sim 10^{-26}$ kg/m ³

4. Physical Significance

Stephan’s Quintet is the most famous compact group collision, extensively studied with JWST Cycle 1 data (2022–2025). The UQFF shock ridge model is unique in connecting the kinetic energy density of the 1500 km/s shock to LENR vacuum coupling — a novel physical mechanism not present in standard hydrodynamic models. The FLEENR term represents the possibility that ultra-high-density shock fronts ($>10^4\times$ ambient) may trigger sub-threshold nuclear reactions mediated by vacuum buoyancy.

The $x_2 = 290$ Mly cosmic baseline sets the scale for UQFF long-range vacuum coherence tests: the $F_{\text{U_Bi_i}} \approx -8.32\times 10^{217}$ N over 290 Mly suggests that UQFF maintains coherent force coupling at intergalactic baselines.

5. Deduplication Note

- **vs. PAPER_346, PAPER_347:** Those are AGN jet systems; this paper is an intergalactic shock ridge (no central BH jet).
- **vs. PAPER_351 (ASASSN-14li):** Both include F_{Kozima} ; ASASSN-14li is a TDE (stellar scale), Stephan’s Quintet is intergalactic.

6. Classification

Physics Territory: FIRST UQFF intergalactic shock ridge with KE_den and LENR coupling

Scale: Intergalactic (290 Mly)

CP Implementation: StephansQuintetShockRidgeFUBiCalculator (Condensed-Physics3.py, Session 96)